Basin Escape: Sufficient Conditions, Guarantees, and Limits

A six-lens mathematical assessment of Player-0 self-settlement, Eureka, and IAE

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Status and provenance. This note extends the supplied nine-page *Basin Escape, Player-0 Activation, and Settlement Preorders* (16 October 2026) [1]. "Six-lens" means six analytical checks inspired by the FORGE-6 discussion, *not* independently running six automated theorem provers. The conditional deductions below were reviewed analytically and repaired for their stated hypotheses; they have not been independently machine-checked and are not experimental speedup measurements. No proving experiment or federation activation was initiated. Version 0.2 preserves the original v0.1 as a separate historical artifact.

Abstract

A basin is a region in which a specified policy makes little certified progress, not necessarily an inaccessible part of the proof space. For fixed target, verifier, legal transition rules, and cost budget, we distinguish leaving a basin from settling a goal and characterize both by admissible finite witnesses; supply stronger, constructive sufficient conditions based on ranking functions, certified lemmas, fair proof enumeration, bounded self-settlement executions, and adaptive search; and separate route existence from execution and speedup. We also state game-theoretic sufficient conditions for forced wins, conditional guarantees for shared BANKs and simulation preorders, and counterexamples to common overclaims.

1 Frozen framework: three different claims

Fix a theory T, target c, sound certificate checker V, initial executable state s_0, a legal transition system $\mathcal{A}$, charged budget b, and a protocol defining accepted success receipts. An execution state includes the search frontier, currently certified lemmas and their undischarged hypotheses, scheduler, activated agents, provenance, and remaining budget. In particular, *the target and verifier do not change when search policy changes.* Write $s \to_{\mathcal{A}} t$ for a legal transition, $\mathsf{cost}(s,t) \geq 0$ for its fully charged cost, and G for states containing a checker-accepted receipt for $T \vdash c$ (or, if explicitly authorized, $T \vdash \neg c$ in an object-level settlement task).

Definition 1.1 (Operational basin and its boundary)**.** *For a policy π, progress functional p and diagnostic window h, a set $B \subseteq S \setminus G$ containing s_0 is a diagnosed search basin if a prespecified observation over the declared number of steps or charged work detects negligible* certified *progress under π. The diagnosis does not assert closure of B under every action of $\mathcal{A}$. A* basin exit *is the first transition from B to $S \setminus B$;* settlement *is a reached state in G. A certified settlement from $s_0 \in B$ necessarily entails a basin exit, but an exit need not settle the goal.*

Definition 1.2 (Possibility, guarantee, and cost)**.** *Relative to $(s_0, B, \mathcal{A}, G, b)$: (i)* possible basin exit *means some finite admissible execution reaches $S \setminus B$ by charge b; (ii)* possible settlement *means some such execution reaches G; (iii) a* guarantee *requires an identified policy that realizes the event on every permitted execution (or with a clearly stated probability); and (iv)* cost improvement *requires a fully charged matched comparator. Mere earlier wall-clock completion does not establish lower total work.*

Theorem 1.3 (Finite witnesses for basin exit and settlement)**.** *Assume $s_0 \in B \subseteq S \setminus G$. Basin exit within budget b is possible if and only if there is a finite legal path from s_0 whose final state lies outside*

B with total charge at most b. Settlement within budget b is possible if and only if there is such a path ending in G:

$$s_0 = s^{(0)} \to_{\mathcal{A}} \cdots \to_{\mathcal{A}} s^{(m)} \in G, \qquad \sum_{i=0}^{m-1} \mathsf{cost}(s^{(i)}, s^{(i+1)}) \le b.$$

Every settlement witness contains a basin-exit step, but the converse need not hold. Neither witness condition alone guarantees that a given search policy will discover the path.

Proof. Each equivalence unpacks the definition of a successful finite execution for the respective destination set. Since $B \cap G = \varnothing$ and $s_0 \in B$, every settlement path first leaves B. A legal exit may lead to a state outside both B and G, showing that exit alone is insufficient. □

Theorem 1.4 (Finite-descent sufficient condition for guaranteed escape)**.** *Suppose an authorized deterministic policy π has a rank $r : S_\pi \to \mathbb{N}$ on all its reachable states, with $r(s) = 0$ implying $s \in G$, and that the policy halts successfully on reaching G. Suppose whenever $s \notin G$ (hence $r(s) > 0$), π takes a legal transition $s \to t$ with $r(t) < r(s)$ and charge at most C (including all obligatory verification, activation, and transfer overhead). Then π reaches G in at most $r(s_0)$ transitions and charge $r(s_0)C$. In particular, $b \ge r(s_0)C$ suffices.*

Proof. The nonnegative integer rank decreases strictly at each nonglobal step, hence there are at most $r(s_0)$ such steps. After at most $r(s_0)$ transitions, the policy must reach G: otherwise another strict decrease would be required below zero. A rank-zero state is in G by assumption. Summing the stepwise cost bound proves the charge bound. □

Remark 1.5. *The rank need not be computable or available in practice; discovering a sound rank and implementing a policy that decreases it is substantive. An empirical "progress score" that sometimes rises cannot serve as this proof without an amended argument.*

2 Method A: Player-0 self-settlement ("White wins")

The supplied note defines the first-person claim as the actual bounded event

$$\mathsf{Win}_b(M, T, c, \mathsf{V}, \xi) = 1 \iff M(\xi) \text{ emits a complete accepted proof of } T \vdash c \text{ by } b.$$

The designated speaker M and frozen input ξ matter. An activated subsidiary's achievement alone does not certify a prediction about a different speaker. The sentence "I will prove the theorem" is a forecast of an operational event, not a new inference rule and not evidence that the event occurs.

Proposition 2.1 (Sufficient condition for successful A)**.** *Assume (A1) Player 0 is legally authorized to activate a specified executable M as part of the frozen protocol; (A2) the activation and handoff preserve T, c, V, actor identity, and dependency/provenance information; (A3) the activated search, together with its authorized caller where applicable, is guaranteed to execute a finite, legally checked proof-producing trace; and (A4) the entire trace, including activation and checking, fits budget b, and the designated M emits the prescribed receipt. Then method A succeeds.*

Proof. Conditions (A1)–(A4) construct an admissible bounded execution satisfying the event's defining receipt condition. □

Remark 2.2 (Noncircular strengthening)**.** *One may discharge (A3) using an independently established rank as in Theorem 1.4, or a finite verified execution schedule, rather than assuming success directly. A proof* about *the future event does not replace the prescribed actual execution. For fixed deterministic finite-time execution with a decidable checker, the resulting bounded event is in principle decidable by faithful execution; the answer still requires execution or an independently sound operational certificate.*

3 Method B: Eureka bridge lemmas

Theorem 3.1 (Certified-bridge composition)**.** *Let H encode the actual frozen admissible hypotheses. Suppose there are finite proof objects π_1 and π_2 accepted respectively for*

$$H \vdash \ell \qquad \text{and} \qquad H, \ell \vdash c,$$

and the proof calculus permits a verified composition (cut/substitution) of these derivations without adding or losing undischarged hypotheses. Then an accepted finite derivation of $H \vdash c$ exists. To settle the frozen objective $T \vdash c$, require additionally that every member of H used by the proof is authorized by T (or established from T), or that all otherwise open assumptions have been validly discharged in the final certificate. Under this requirement, if a specified policy can construct/import both proofs, verify their composition, and emit a $T \vdash c$ receipt within b, settlement by B is guaranteed.

Proof. Substitute π_1 for any undischarged use of ℓ in π_2, or invoke the calculus's checked cut rule. Dependency tracking ensures every surviving assumption belongs to H. The additional discharge or admissibility condition ensures that the final verifier checks the frozen judgement $T \vdash c$, not merely a stronger-hypothesis judgement. The operational statement follows if the declared policy executes this finite construction and fits the budget. □

What makes a bridge useful? The logical relation $H \vdash \ell$ adds no new axioms. The bridge can still change the *search graph*, make a difficult step locally accessible, or permit reuse. A useful nontrivial bridge should be cheaper to discover and exploit than the original search obstacle, actually accessible to the receiving prover, and neither merely the target renamed nor an illicit assumption. This is a design criterion, not a theorem that every valid bridge helps.

Proposition 3.2 (Explicit cost criterion for improved Eureka escape)**.** *Let C_0 be the full cost of a matched baseline success. If B yields a verified target proof at charged total cost*

$$C_B = C_{\text{setup}} + C_{\text{find}\,\ell} + C_{\text{verify}\,\ell} + C_{\text{handoff}} + C_{\text{finish}|\ell} + C_{\text{verify}\,c} + C_{\text{coord}} + C_{\text{retries}},$$

with all displayed categories aggregating all nonoverlapping mandatory costs, then $C_B < C_0$ is sufficient for strict improvement in total charged cost on that instance. This inequality is a comparison criterion, not evidence that it holds for any actual run.

Counterexample 3.3 (A valid but unhelpful lemma)**.** *Let $H = \{p, p \to q\}$, $c = q$ and $\ell = p$. The bridge is valid and permits q in one modus-ponens step, but if the baseline already uses that step, rediscovering and sharing p adds cost rather than reducing it.*

Remark 3.4 (Proof sharing can be exponentially significant)**.** *Proof-complexity literature exhibits exponential separations between tree-like and DAG-like proof systems for some families [4]. Shared certified intermediate derivations provide a concrete mechanism for avoiding repeated subproof construction. This supports* possibility on some families*, not a universal exponential-to-linear conversion or a performance claim for the proposed federation.*

4 Method C: Iterative Adaptive Eureka

A legitimate IAE stage chooses a legal policy or candidate-lemma query using permitted observations. Only independently verified items enter BANK; all trials and checks are charged.

Theorem 4.1 (Fair enumerative discovery)**.** *Assume that the frozen static task admits an effective enumeration of finite proposed proof traces, including every valid finite successful trace; candidate verification terminates; and competing traces can be evaluated in isolated reproducible states (or as self-contained proof certificates) without altering the frozen task. A dovetailed exhaustive procedure that grants each candidate finite prefix unbounded cumulative work eventually detects any existing successful finite trace. If a valid* a priori *upper bound on all required enumeration, reconstruction, and checking charges is b, settlement is guaranteed within b.*

Proof. A successful finite trace has a finite description and thus appears at a finite index of the enumeration. Each of its finitely many required checks receives sufficient work under dovetailing and terminates, so the receipt is eventually identified. Isolation/reproducibility makes parallel candidate testing legal rather than illicitly combining mutually incompatible state histories. The charged upper bound yields the bounded claim. □

Remark 4.2. *Effective completeness gives no uniform finite deadline. It does not decide all non-theorems or prove independence by a failed search, and it does not apply unchanged to moving-adversary games in which one favorable trace is insufficient.*

Theorem 4.3 (Constructive bounded-tree discovery)**.** *Suppose an authorized search policy can exhaust a rooted tree of legal, reproducible static proof attempts of depth at most d, with at most $k \geq 1$ children per node, using breadth-first expansion. Suppose some node at depth at most d carries a verifier-accepted proof of the frozen target, and that complete expansion of one node, including reconstruction, checking, queue management, and receipt generation where applicable, costs at most C. Then this explicit policy discovers a settlement using at most*

$$N_{k,d} = \sum_{j=0}^{d} k^j \quad \textit{node expansions and total charge at most} \quad CN_{k,d}.$$

Thus $b \geq CN_{k,d}$ is a constructive sufficient budget.

Proof. A breadth-first traversal exhausts every node through level d. Bounded branching gives at most $N_{k,d}$ nodes. By hypothesis the successful node is among them, and each expansion terminates with all costs included. Summing the uniform per-node charge proves the claim. □

Remark 4.4. *The finite-tree theorem guarantees discovery rather than postulating that the system will happen to follow a successful path. Its hypotheses can be expensive: $N_{k,d}$ is generally exponential in d for $k > 1$.*

Theorem 4.5 (Probabilistic adaptive escape)**.** *Let E_i be success with a verified receipt on trial i, and take all displayed conditional probabilities over positive-probability prior-failure histories. If the policy preserves an admissible opportunity to try again and, for each reached trial,*

$$\Pr(E_i \mid E_1^c \cap \cdots \cap E_{i-1}^c) \geq p_i, \qquad 0 \leq p_i \leq 1,$$

then

$$\Pr(\textit{success in first } m \textit{ trials}) \geq 1 - \prod_{i=1}^{m}(1 - p_i) \geq 1 - \exp\Big(-\sum_{i=1}^{m} p_i\Big).$$

If $\sum_i p_i = \infty$ and every required further trial remains legally executable, success occurs with probability one. If each complete trial (including its success verification) costs at most C and $p_i \geq p > 0$, the expected aggregate trial charge through first success is at most C/p, plus any separately accounted non-trial setup cost.

Proof. By the chain rule, probability of m consecutive failures is the product of the successive conditional failure probabilities and hence at most $\prod(1-p_i)$. Use $1-x \le e^{-x}$. A divergent probability sum drives the upper bound to zero. With a uniform lower bound p, $\Pr(N > m) \le (1-p)^m$, so $\mathbb{E}[N] \le \sum_{m\ge 0}(1-p)^m = 1/p$; multiply by the per-trial charge bound. Independence is not needed. □

Remark 4.6. *The theorem's p_i bounds are assumptions to establish, not inferred from observing "stagnation." The finite-budget guarantee is probabilistic unless a deterministic successful schedule is shown. Adaptive feedback can be uninformative or adversarial; a policy that permanently abandons the only successful method breaks fair discovery.*

5 Combined A+B+C: sufficient conditions

Theorem 5.1 (Federated certified-escape theorem)**.** *Suppose (F1) a frozen target c, a theory T, an independent sound checker V, and a recorded relation between H and T that allows every final open assumption to be checked or discharged as a proof of $T \vdash c$; (F2) permitted Player-0 activation and a well-defined designated speaker; (F3) a finite chain of lemmas $\ell_1, \dots, \ell_k$ with checked derivations whose open assumptions are discharged solely by H or earlier eligible lemmas; (F4) a checked terminal derivation of c from H and this chain; (F5) a specified scheduler that either satisfies Theorem 1.4 for successful goal states or exhaustively explores a reproducible finite search tree satisfying Theorem 4.3, whose candidate space includes that chain and terminal proof; (F6) every BANK handoff preserves proof dependencies and target identity; and (F7) the appropriate rank or tree bound, including all necessary search, checking, activation, communication, and receipt costs, is at most b. Then the designated federation settles c by b. It also satisfies the first-person prediction precisely when the frozen protocol designates that federation as speaker (or legally assigns its output to the specified speaker) and that speaker emits the required receipt.*

Proof. Induct over the bridge chain. The first lemma is certified under H; if the first $i-1$ lemmas are available with proper dependencies, the ith verified derivation is safely composable. Compose the terminal proof, discharging every assumption not permitted by the frozen judgement $T \vdash c$ as required by (F1). The certified rank policy or finite-tree exhaustive scheduler in (F5), together with (F7), ensures that this legal certificate is actually reached and emitted within budget. The protocol-bound actor and receipt requirement gives the last clause. □

Corollary 5.2 (Optional extensions preserve possibility)**.** *If an extended federation retains every legal old action path unchanged at the same charge, without mandatory overhead, then adding A, B, and C cannot remove any formerly possible escape. This does not assert that a new scheduler will* choose *an old winning path or that costs improve.*

6 Opponents, static proofs, and adaptive targets

6.1 Chess and other adversarial games

A verified favorable chess line covers only one possible Black response sequence; a forced win requires $\exists\sigma_W\,\forall\sigma_B$. For a fixed finite-horizon turn-based game, let W_0 be declared White-winning terminal states and recursively define

$$W_{k+1} = W_k \cup \{s \text{ (White turn)} : \exists(s \to t), t \in W_k\} \cup \{s \text{ (Black turn)} : \forall(s \to t), t \in W_k\}.$$

The displayed recurrence applies only to nonterminal positions with at least one legal move; terminal/no-move conventions must be specified separately. If the initial rule-complete state lies in W_k and the

legal transition relation and terminal classifications are correct, induction gives a White strategy forcing the target within at most k individual legal moves (plies); the recurrence is not counting full pairs of turns. That is a sufficient condition for *game-theoretic* basin escape, different from the first-person operational proof event.

6.2 Static proofs and password tasks

In theorem proving, the adversary does not move the frozen T or c; counterexample/refutation branches test the same problem. For a *fixed* finite password candidate set of size N known to contain the fixed secret, if equality tests are sound, every candidate may legally be tested, the target remains fixed, and the total charged cost of up to N tests fits the budget, exhaustive search guarantees success. If the secret or acceptance rule may adapt without restrictions, that guarantee disappears; a bound on adaptation or a winning-strategy argument is then needed.

7 Six analytic audits (not six executed solvers)

These are six analytical perspectives designated B_0–B_5 for this report, *not* claimed outputs of six actual FORGE-6 ATP executions and not a modification of the historical AMLD+0 roles.

- B_0 **Legality and actor identity:** Are activation, schedules, recursion, delegation, and speaker/receipt identity actually authorized? Without them A may be only a sentence.
- B_1 **Proof validity:** Does a complete accepted $T \vdash c$ proof path exist, with all extra assumptions discharged, and will the selected policy execute it rather than merely predict it?
- B_2 **Bridge construction:** Are intermediate lemmas nontrivial, certifiable, composable, and genuinely accessible to the receiving prover?
- B_3 **Adaptive completeness:** Does IAE allocate recurring opportunities to productive alternatives, or can premature abandonment cause permanent stagnation?
- B_4 **Obstructions and adversaries:** Is a purported exit blocked by a legal-action invariant, exhausted budget, an independent target, or unmodelled adversarial responses?
- B_5 **Dependencies and full cost:** Does every shared entry retain all undischarged assumptions, and are verification, transport, retries, and coordination included in reported improvements?

These six perspectives do not certify logical independence, imply a universal ATP, or establish experimental execution.

8 Boundary conditions and impossibility witnesses

Proposition 8.1 (Closed-obstruction condition)**.** *If $B \cap G = \varnothing$ and all legally reachable successors of every $s \in B$ lie in B, then no protocol restricted to those legal transitions can escape B. Likewise, no path can satisfy a cost budget smaller than the infimum charge of legal successful paths (where a path exists).*

Proof. Closure inductively keeps every finite execution in B, disjoint from G. The cost claim follows from the definition of an infimum. □

Counterexample 8.2 (Possibility without discovery)**.** *A state s has a zero-progress self-loop and one legal exit to a state with an immediate accepted proof. A deterministic policy that always chooses the self-loop never escapes, although a two-step successful path exists. Merely adding an exit or telling the policy "you will win" proves no operational success.*

Counterexample 8.3 (A stronger-hypothesis proof is not target settlement)**.** *Take $T = \varnothing$ and $H = \{c\}$, where c is not a logical validity. Then $H \vdash c$ has an immediate derivation, but $T \nvdash c$. A Eureka bridge built from unlicensed H does not settle the frozen T-goal.*

Counterexample 8.4 (Sharing without a sound BANK)**.** *An agent certifies $T \cup \{a\} \vdash \ell$ and shares ℓ as if $T \vdash \ell$. A receiving prover may then appear to prove c from T using an unproved assumption. This is not a legal escape even if a permissive logger accepts the text.*

Remark 8.5 (Limit of global guarantees)**.** *For general effectively presented first-order theories, semidecision of provability by proof enumeration is possible under standard enumerability assumptions; universal decision of* both *theoremhood and non-theoremhood is not thereby obtained. Unbounded search cannot be advertised as a uniform deadline or a proof of independence on failure. The source note's SIC upward closure holds only for certificate-preserving legal simulations under the applicable cost transformation; well-foundedness is additionally needed for minimal-antichain generation [1].*

9 Evidence, implications, and a concise sufficient-condition checklist

Peer-reviewed proof-complexity results show rigorous tree/DAG separations [4]; research on restarts establishes that controlled randomization can improve difficult search distributions on some families but does not make every hard problem easy [5]; distributed SAT research shows that independently verifiable proof reconstruction from shared work is implementable [2], with newer parallel proof representations demonstrating further scalability [3]. These are *external analogues*, not measurements of AMLD+0, Eureka, IAE, or FORGE-6.

Seven sufficient-condition questions for any proposed escape:

- **S1.** Is the frozen theorem, hypothesis set, designated speaker, checker, and accepted goal explicit?
- **S2.** Is there a finite legal path (or winning game strategy) to that goal?
- **S3.** Are intermediate products proved, independently verified, and free of unauthorized open assumptions?
- **S4.** Does the chosen mechanism actually provide an authorized transition out of the diagnosed basin?
- **S5.** Will a specified policy execute/discover the complete route (via rank, completeness/fairness, or a quantified success probability)?
- **S6.** Does the complete route fit the budget after all charges, including checking and delegation?
- **S7.** Does the correct actor emit the exact accepted certificate/receipt, and is any speedup assessed only relative to a fully charged baseline?

Final judgment. The strongest defensible claim is a *conditional possibility and guarantee theorem*: verified bridges, legal activation, certified composition, a specifically terminating rank policy or complete bounded-tree schedule, and sufficient fully charged resources suffice for success. The source note does *not* establish that every basin has such a bridge, that any particular proposed agent can find one, or that a federation is universally faster. The next genuinely new theorem would identify a nontrivial task family together with a constructive rank/fairness argument and an audited asymptotic (or empirical) cost bound.

References

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